Multilevel Modeling for Longitudinal Data

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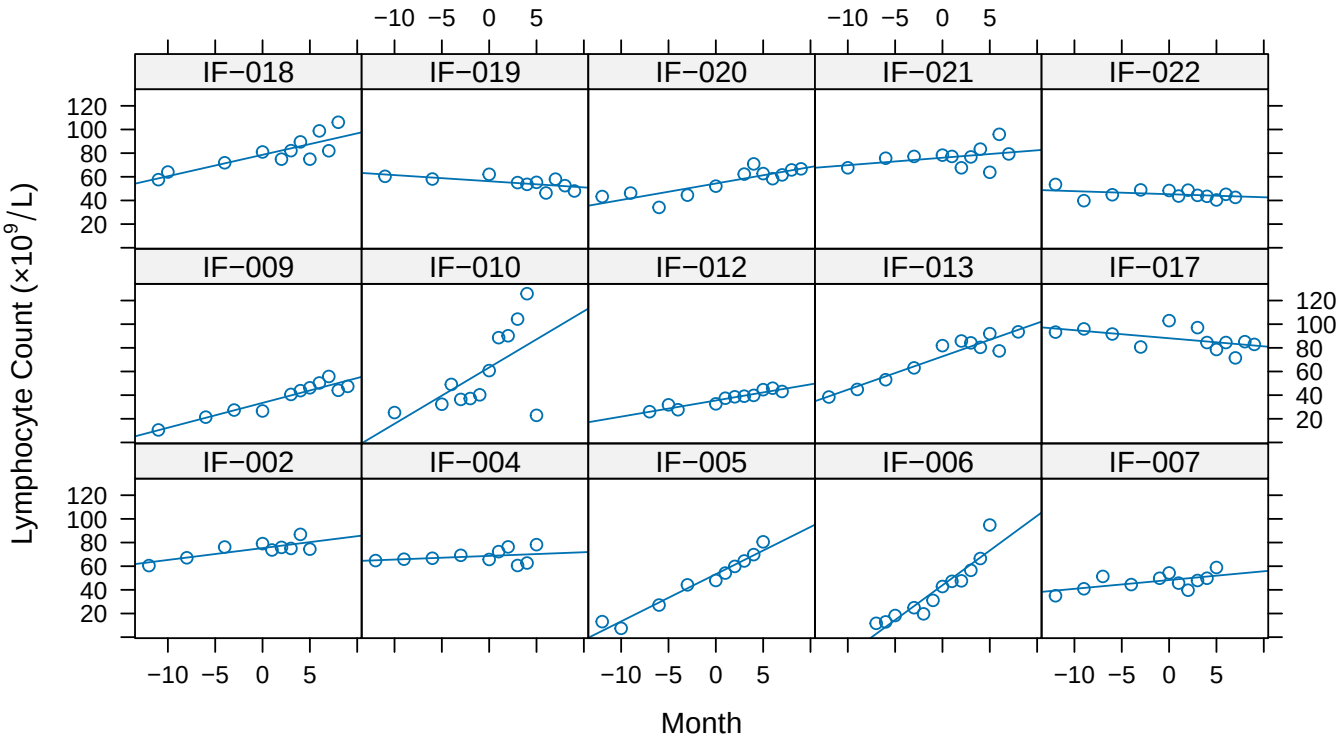
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1. Data description

The longitudinal data are from an experiment to study the effect of intermittent fasting (IF) on the lymphocyte measurements of 15 participants, each of whom was repeatedly measured before, during, and after the intervention. The data set (long format) consists of six columns:

- **LYMP** - [numeric] lymphocyte count ($\times 10^9$/L); lower values usually correspond to better outcomes;
- **ATG** - [numeric] relative net LC3-II expression; higher levels are associated with increased autophagy;
- **ID** - [factor] participant identifier; 15 levels;
- **Time** - [numeric] time point, with a frequency of every 30 days, treating the last pretest as Month 0;
- **Time_Within_Phase** - [numeric] time point within each phase, starting from Month 0;
- **Phase** - [numeric] treatment identifier; 0 as pretest, 1 as intervention, and 2 as posttest.

```
df <- read.csv("~/Downloads/IF_v2.csv", stringsAsFactors=T)
lattice::xyplot(LYMP ~ Time | ID, data=df, type=c("p","r"),
                xlab="Month", ylab=expression(paste("Lymphocyte Count (\u00d7",10^9/L,")")))
```



2. Linear mixed-effect regression (Part I)

The hypothesis is that IF can help patients lower their lymphocyte count. The objectives are then to investigate whether (1) the mean responses and (2) the linear slopes of time are statistically different before versus during the intervention.

2.1 Using the "nlme" package (v 3.1-162)

```
dat <- df[!is.na(df$LYMP) & df$Phase!=2,] #IF-013 contains one missing value

fit1 <- nlme::lme(fixed=LYMP ~ factor(Phase),
                  random=~1 | ID,         #random intercept for each subject
                  data=dat, method="ML")
summary(fit1)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1224.717 1236.706 -608.3586
##
## Random effects:
##  Formula: ~1 | ID
##         (Intercept) Residual
## StdDev:    15.96643 12.81501
##
## Fixed effects:  LYMP ~ factor(Phase)
##                   Value Std.Error  DF   t-value p-value
## (Intercept)    50.17287  4.422792 132 11.344163       0
## factor(Phase)1 14.12315  2.161386 132  6.534303       0
##  Correlation:
##                (Intr)
## factor(Phase)1 -0.246
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -2.49165610 -0.55618518 -0.03192352  0.62292969  4.25175278
##
## Number of Observations: 148
## Number of Groups: 15
```

`~1 | ID` is the random effects part of the model, allowing for the possibility that different subjects have different baseline lymphocyte counts. `LYMP ~ factor(Phase)` specifies the fixed effect, expecting the dependent variable to be influenced by the categorical treatment identifier.

This model (`fit1`) compares the dependent variable (`LYMP`) between the different phases of the experiment (`Phase`), while controlling for the repeated measures within each subject.

Two methods are available to estimate the coefficients: the restricted maximum likelihood (REML) by default and the maximum likelihood (ML). In a nutshell, ML is generally preferred when comparing models with different fixed effects.

```r
fit2 <- nlme::lme(fixed=LYMP ~ factor(Phase) * Time,
                  random=~Time | ID,      #random intercept and slope for each subject
                  data=dat, method="ML")
summary(fit2)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##       AIC      BIC    logLik
##   1124.47 1148.447 -554.2349
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.522600 (Intr)
## Time         2.019187 -0.089
## Residual     7.148936
##
## Fixed effects:  LYMP ~ factor(Phase) * Time
##                        Value Std.Error  DF   t-value p-value
## (Intercept)         57.89561  4.558001 130 12.701974  0.0000
## factor(Phase)1       3.71169  2.519625 130  1.473110  0.1431
## Time                 1.49179  0.573226 130  2.602454  0.0103
## factor(Phase)1:Time -0.13552  0.579038 130 -0.234052  0.8153
##  Correlation:
##                     (Intr) fc(P)1 Time
## factor(Phase)1      -0.177
## Time                 0.019 -0.178
## factor(Phase)1:Time -0.100 -0.514 -0.144
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -3.26424016 -0.35417230 -0.04107843  0.49755880  3.70258899
##
## Number of Observations: 148
## Number of Groups: 15
```

`~Time | ID` adds random slopes for the time variable, which will allow the effect of time to vary across subjects. The interaction term `factor(Phase) * Time` allows the slope of the time trend to differ between the phases.

If the slope of the time trend is the same across phases, one can remove the interaction term and specify `factor(Phase) + Time` as fixed effects.

2.2 Using the "lme4" package (v 1.1-33)

```
fit1 <- lme4::lmer(LYMP ~ factor(Phase) + (1 | ID), dat, REML=FALSE)
summary(fit1)

fit2 <- lme4::lmer(LYMP ~ factor(Phase) * Time + (Time | ID), dat, REML=FALSE)
summary(fit2)

fit3 <- lme4::lmer(LYMP ~ factor(Phase) + Time + (Time | ID), dat, REML=FALSE)
summary(fit3) #(same) output omitted
```

In linear mixed-effect regression (LMER), each subject has its intercept and slope (i.e., random intercepts and random slopes).

The p-value of .8153 for the interaction term in `fit2` is not statistically significant at an alpha level of .05 ($t = -0.234052$, $df = 130$). Hence, fail to reject that the linear slopes of time are the same before versus during the intervention.

▶ In summary, the mean lymphocyte count continued to increase after participants received IF treatment.

```
fit3 <- nlme::lme(fixed=LYMP ~ factor(Phase) + Time,
                  random=~Time | ID,
                  data=dat, method="ML")
summary(fit3)
```

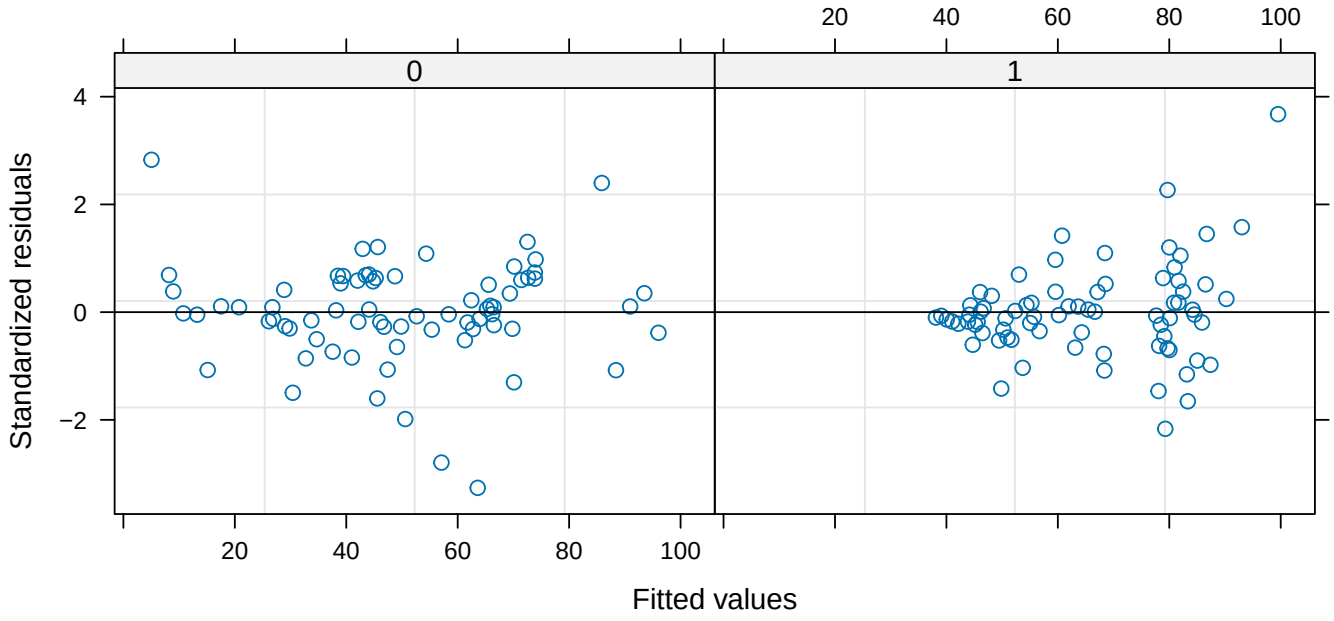
```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1122.526 1143.506 -554.2628
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.508068 (Intr)
## Time         2.026421 -0.087
## Residual     7.148179
##
## Fixed effects:  LYMP ~ factor(Phase) + Time
##                   Value Std.Error  DF   t-value p-value
## (Intercept)    57.79047  4.515995 131 12.796839  0.0000
## factor(Phase)1  3.40709  2.152902 131  1.582559  0.1159
## Time            1.47275  0.567068 131  2.597142  0.0105
##  Correlation:
##                (Intr) fc(P)1
## factor(Phase)1 -0.267
## Time            0.007 -0.296
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -3.2625501 -0.3584044 -0.0429246  0.5105937  3.6754491
##
## Number of Observations: 148
## Number of Groups: 15
```

3. Diagnostics

3.1 Check for linearity and homoscedasticity

Plot standardized residuals versus fitted values by phase. One should see a random scattering of points if the assumptions are met. Plus, check for influential outliers.

```
plot(fit3, resid(., type = "p") ~ fitted(.) | factor(Phase), abline=0)
```



The intervention phase (right panel, `Phase==1`) may witness heteroscedasticity according to the residual plot. Using the `weights` argument, one can incorporate a variance function structure into the model. The variance structure `varIdent(form=~1 | factor(Phase))` allows for different variances for each phase level.

```
fit4 <- nlme::lme(fixed=LYMP ~ factor(Phase) + Time,
                  random=~Time | ID,
                  cor=corCompSymm(),
                  weights=varIdent(form=~1 | factor(Phase)),
                  data=dat, method="ML")
summary(fit4)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1116.744 1143.719 -549.3719
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 18.654109 (Intr)
## Time         1.740448 0.701
## Residual    24.150594
```

```
##
## Correlation Structure: Compound symmetry
##  Formula: ~1 | ID
##  Parameter estimate(s):
##       Rho
## 0.8934861
## Variance function:
##  Structure: Different standard deviations per stratum
##  Formula: ~1 | factor(Phase)
##  Parameter estimates:
##         0         1
## 1.0000000 0.6887895
## Fixed effects:  LYMP ~ factor(Phase) + Time
##                   Value Std.Error  DF  t-value p-value
## (Intercept)    58.15180  7.825701 131 7.430874  0.0000
## factor(Phase)1  2.96753  2.848364 131 1.041837  0.2994
## Time            1.46597  0.500614 131 2.928351  0.0040
##  Correlation:
##                 (Intr) fc(P)1
## factor(Phase)1 -0.625
## Time            0.454 -0.269
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -2.6474784 -0.5986648  0.1343088  0.6488575  2.1497415
##
## Number of Observations: 148
## Number of Groups: 15
```

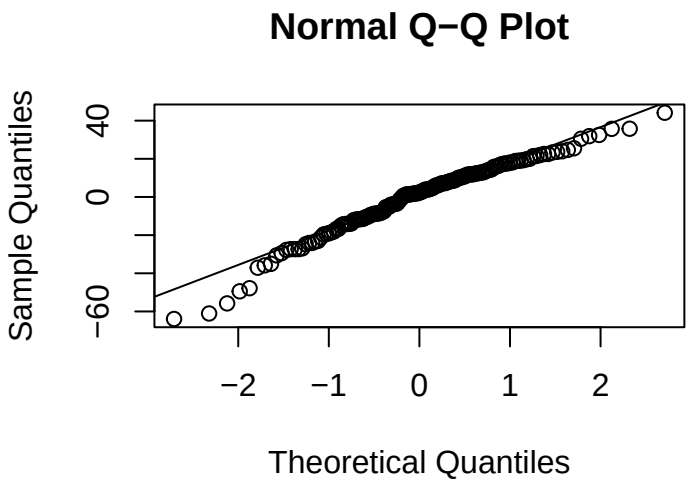
```
coef(fit4)
```

```
##        (Intercept) factor(Phase)1       Time
## IF-002    59.32070        2.96753  1.4111258
## IF-004    55.62287        2.96753  0.2255899
## IF-005    62.31410        2.96753  3.0766349
## IF-006    62.14011        2.96753  3.3017775
## IF-007    38.40942        2.96753  0.3976132
## IF-009    49.97750        2.96753  1.4889376
## IF-010   110.22773        2.96753  6.0195517
## IF-012    43.41974        2.96753  0.4736706
## IF-013    69.84448        2.96753  2.9312904
## IF-017    58.85621        2.96753 -0.1613294
## IF-018    61.64983        2.96753  1.5173742
## IF-019    44.07036        2.96753 -0.4114820
## IF-020    59.57141        2.96753  1.0633564
## IF-021    56.31475        2.96753  1.0211253
## IF-022    40.53781        2.96753 -0.3656464
```

3.2 Check for normality of the residuals

The residuals should follow a straight line in a QQ plot (Quantile-Quantile plot) if normally distributed.

```
qqnorm(resid(fit4)); qqline(resid(fit4))
```

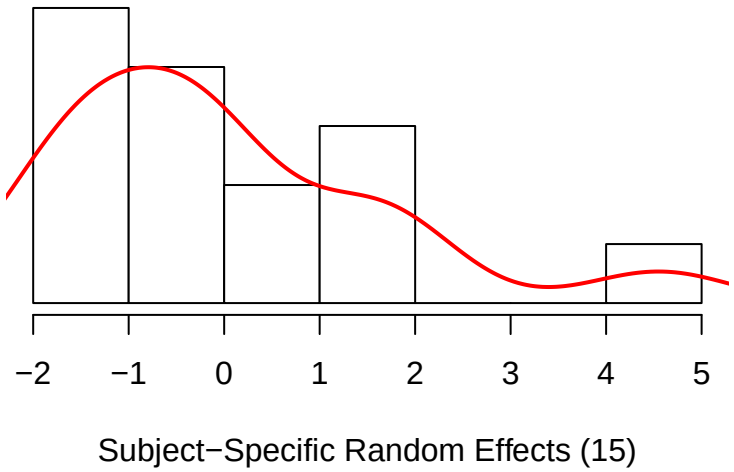


3.3 Check for random effects assumptions

Check that the random effects are normally distributed and have a constant variance. One way to do this is to extract the random effects from the model and plot histograms.

```
rand.ef <- nlme::ranef(fit4)$Time #15 subjects

hist(rand.ef, prob=TRUE, col="white", yaxt="n",
     ylab="", xlab="Subject-Specific Random Effects (15)", main="")
lines(density(rand.ef),col="red",lwd=2)
```



3.4 Check for multicollinearity

Generally, a variance inflation factor (VIF) value greater than 5 could indicate a multicollinearity problem.

```
#Note that the vif() function in the "car" R package may not work directly with lme objects.
car::vif(lm(LYMP ~ factor(Phase) + Time, data=dat))
```

```
## factor(Phase)          Time
##      3.037482      3.037482
```

4. Linear mixed-effect regression (Part II)

Investigate (3) any difference in either autophagy or lymphocyte count between those who fasted for three versus six months.

```
lymphocyte <- df[df$Phase==1, -2]
autophagy <- df[!is.na(df$ATG) & df$Phase==1, -1] #Excluded IF-007 and 012
fast6 <- paste0("IF-0",c(17,19,20,21,22))
lymphocyte$Fast <- factor(ifelse(lymphocyte$ID %in% fast6, "Six", "Three"))
autophagy$Fast <- factor(ifelse(autophagy$ID %in% fast6, "Six", "Three"))

fit.LYMP <- nlme::lme(fixed=LYMP ~ Fast * Time_Within_Phase,
                      random=~Time_Within_Phase | ID,
                      data=lymphocyte, method="ML")
summary(fit.LYMP)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: lymphocyte
##        AIC      BIC    logLik
##   564.0117 582.5516 -274.0059
##
## Random effects:
##  Formula: ~Time_Within_Phase | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##                    StdDev    Corr
## (Intercept)       16.516612 (Intr)
## Time_Within_Phase  1.528240 0.432
## Residual           6.191262
##
## Fixed effects:  LYMP ~ Fast * Time_Within_Phase
##                                 Value Std.Error DF   t-value p-value
## (Intercept)                 65.07029  7.835393 58  8.304662  0.0000
## FastThree                   -3.84529  9.645958 13 -0.398642  0.6966
## Time_Within_Phase           -0.22457  0.884671 58 -0.253848  0.8005
## FastThree:Time_Within_Phase  3.42457  1.356162 58  2.525193  0.0143
##  Correlation:
##                             (Intr) FstThr Tm_W_P
## FastThree                   -0.812
## Time_Within_Phase            0.207 -0.168
## FastThree:Time_Within_Phase -0.135  0.105 -0.652
##
```

```
## Standardized Within-Group Residuals:
##          Min           Q1          Med           Q3          Max
## -2.417873860 -0.422707000  0.007624772  0.396155990  2.584876669
##
## Number of Observations: 75
## Number of Groups: 15
```

```
fit.ATG <- nlme::lme(fixed=ATG ~ Fast * Time_Within_Phase,
                     random=~Time_Within_Phase | ID,
                     data=autophagy, method="ML")
summary(fit.ATG)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: autophagy
##         AIC      BIC   logLik
##   -5.651109 11.23588 10.82555
##
## Random effects:
##  Formula: ~Time_Within_Phase | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##                    StdDev     Corr
## (Intercept)       0.30856306 (Intr)
## Time_Within_Phase 0.05083683 -0.917
## Residual          0.15131001
##
## Fixed effects:  ATG ~ Fast * Time_Within_Phase
##                                    Value  Std.Error DF   t-value p-value
## (Intercept)                    0.6166054 0.15183125 46  4.061123  0.0002
## FastThree                      0.0722235 0.19562025 11  0.369203  0.7190
## Time_Within_Phase             -0.0312958 0.02767315 46 -1.130910  0.2640
## FastThree:Time_Within_Phase -0.0722687 0.04231507 46 -1.707871  0.0944
##  Correlation:
##                               (Intr) FstThr Tm_W_P
## FastThree                     -0.776
## Time_Within_Phase             -0.882  0.685
## FastThree:Time_Within_Phase    0.577 -0.808 -0.654
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -1.5396561 -0.5850891 -0.1483048  0.5270094  2.6615711
##
## Number of Observations: 61
## Number of Groups: 13
```

► The p-value of .0143 for the interaction term in `fit.LYMP` is statistically significant at an alpha level of .05 ($t = 2.525193$, $df = 58$). Hence, reject that the linear slopes of time are the same between the two fasting cohorts. Participants who fasted for three months showed a higher lymphocyte growth rate than those who fasted for six months.

► The p-value of .0944 for the interaction term in `fit.ATG` is not statistically significant at an alpha level of .05 ($t = -1.707871$, $df = 46$). Hence, fail to reject that the linear slopes of time are the same between the two fasting cohorts. There was no difference between the autophagy values of fasting for three and six months.